MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR

EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - MG1MDCGEO100 - UNDERSTAND THE EARTH

(2024 ADMISSION ONWARDS)

QP CODE: 24900121 **ANSWER KEY**

Duration: 1 Hours Maximum Marks: 35

(Short notes on any 7 out of 8 ; 2 marks each)

1. Polar diameter = 12,714 (1 mark) Equatorial diameter=12,756 kilometers (1 mark) [U] [1]
2. Polar radius = 6357 km equatorial radius = 6378 km (1 mark) Absolute difference 21 km (1 mark) [U] [1]
3. Sedimentary petrology- study of sedimentary rocks-formed from sediments generated by weathering and erosion (1 mark) Igneous petrology – study of igneous rocks – formed from magma or lava (1 mark) [U] [2]
4. Petroleum is a fossils fuel having bulk chemistry – Hydrocarbons, Nitrogen, Oxygen and Sulfur. (1 mark) (1 mark) [K] [3]
5. Endogenous – Processes operates within the Earth. Exogenous - processes on Earth's surface (1.5 marks). Examples (0.5 marks) [U] [3]
6. Any four measures- Avoid slope modifications, Planting deep rooted trees, Not to provide barriers for surface water flow, Protecting failed slopes by constructing retaining walls (0.5 each) [K] [4]
7. Transform –lithospheric plate movement in which plates slide past each other (1 mark) Divergent – plates move away from each other (1 mark) [U] [3]
8. Weathering – erosion/ transportation- deposition- consolidation/ lithification. (1 mark) [U] [2]

$(7 \times 2 = 14)$

Part B

Write Short Notes on Any 3 of the following In 230 Words

Each question carries 7 marks

9. Discuss the dimensional characteristics of the Earth [U] [1]
   Earth is a terrestrial planet/ or an introduction (2 mark). Size- 5th largest planet, equatorial radius is 6,378.14 km, polar radius is 6,356.75 km (2 marks). Shape- 1 mark - Oblate spheroid/ Geoid. Volume -1 mark- 260 billion cubic miles. Density- 1 mark - 5.5gm/cm3.

10. Describe fossil fuel, list the types of fossil fuels and explain the process of its formation. [K] [3]
    Petroleum, coal, and natural gas are called fossil fuels -they formed from the remains of plants and animals (1 mark)

Coal is an organic sedimentary rock made up of decomposed and compacted plant remains. When plants die, their remains usually decompose by reaction with oxygen. In a reducing environment, undecayed or partially decayed plant remains form peat. As peat is buried and compacted by overlying sediments, it converts to coal, a hard, black, combustible rock, Plant matter is composed mainly of carbon, hydro gen, and oxygen and contains large amounts of water. During burial, rising pressure expels the water and chemical reactions release most of the hydrogen and oxygen, and the proportion of carbon increases. The result is coal, a combustible rock composed mainly of carbon (3 marks).

Petroleum /crude oil is a fossil fuel having chemistry -HC+ NSO compounds. Streams carry organic matter from decaying land plants and animals to the sea and to some large lakes, and deposit it with mud in shallow coastal waters. Marine plants and animals die and settle to the sea floor, adding more organic matter to the mud. Younger sediment then buries this organic-rich mud. Rising temperature and pressure resulting from burial convert the mud to shale. At the same time, the elevated temperature and pressure convert the organic matter to liquid petroleum that is finely dispersed in the rock. The activity of bacteria may enhance the process. Typically, petroleum forms in the temperature range from 50 to 100ºC. At temperatures above about 100ºC, oil begins to convert to natural gas (3 marks).

11. Compare and contrast the three rock types- igneous, sedimentary and metamorphic rocks [U] [2]
Igneous rock – primary rocks formed from magma or lava, Extrusive and intrusive igneous rocks with examples List the major rock forming minerals in an igneous rock- for example -major minerals in granite (2marks)
Sedimentary rocks- definition Their origin involves the weathering, transportation of the sediments and their deposition in a sedimentary environment, followed by compaction and cementation, characterized by layered nature (bedding) Type of sedimentary rocks (2 mark). Metamorphic rocks are formed from igneous or sedimentary rocks under the influence of change in pressure and temperature conditions, and chemically active fluids under sub-surface condition. Cite two examples (2 marks). Compare the characteristics of rock types (1 mark).

12. Discuss institutional framework and policies to manage disasters in national and state level [U] [4]

After the disaster occurrence the primary responsibility for undertaking the rescue, relief and rehabilitation measures are to be undertaken by State Government. Central Government supports with financial and physical resources. Disaster Management Act -2005-Proposes Three tier institutional structure –National, State and District Levels. NDMA: National Disaster Management Authority; SDMA: State Disaster Management Authority; DDMA: District Disaster Management Authority (4 marks)
National Crisis Management Committee (NCMC) - Command, Control and Coordination of disaster response under the Cabinet Secretary. NCMC issue guidelines from time to time.
Ministry of Home Affairs (MHA)-Nodal agency at National level of coordination of response and relief. Povide Financial and logistic support to the State Govt.
National Executive Committee (NEC)- C under the Union Home Secretary will coordinate response in the event of disaster. NEC provides guidance to the measures to be taken by the Ministries/Departments at the State and National level.

State Executive Committee (SEC)-State hasthe primary responsibility to respond to disaster. State Chief Secretary have to coordinate.

Department of Relief and Disaster management –Nodal Dept. Secretary of the Dept./Relief Commissioner shall implement the decisions of SEC.Other Departments of the State provide emergency support. (3 marks)

13 Classify plate boundaries with neat sketches [U] [3]

Plate boundaries are marked by tectonic activities (Definition 1 mark.) Divergent plate boundaries are zones where the plates split and spread apart-Processes at this plate margin with diagram (2 marks). Convergent plate boundaries are zones where plates meet. Processes at this plate margin with diagram (2 marks). Transform fault boundaries are zones where plates slide horizontally past each other. Processes at this plate margin with diagram (2 marks).

(3 × 7 = 21 marks)

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